LLOYD GOMEZ

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EDUCATION

The University of Texas at Austin May 2029
B.S. Electrical & Computer Engineering | GPA: 3.91
Electronics and Integrated Circuits

TECHNICAL SKILLS

Programming: Verilog, C++, C, Assembly Python, Java, CSS
Engineering: KiCad, Revit, Altium, LTspice, ARM, x86, Laser cutting | Tools: Microsoft Excel and Word

PROJECTS

Levitation Control PCB – *Texas Guadaloop* August 2026 - October 2026
- Engineered a 60 V PWM-controlled H-bridge to reverse and modulate current through the pod's levitation coil
- Implemented bidirectional current feedback so the embedded controller can regulate coil current
- Enabled reversible coil drive to adjust lift around permanent-magnet bias in the pod's hybrid suspension

Medium Voltage PCB – *Texas Guadaloop* May 2026 - July 2026
- Designed a 36-72 V-to-24 V synchronous converter targeting 20 A for the pod's auxiliary power supply
- Configured soft start, undervoltage lockout, and current limiting to manage startup and overloads
- Distributed a common regulated supply through four 24 V branches to power the pod's auxiliary subsystems

High Voltage Capacitor Monitoring PCB – *Texas Guadaloop* February 2026 - May 2026
- Designed isolated 600 V capacitor sensing to translate the pod's HV bus voltage into a low-voltage signal
- Implemented analog detection targeting 60 V to enable a flashing warning light without embedded firmware
- Integrated galvanic isolation and 24 V alarm drive to indicate residual capacitor charge during pod servicing

Buck Converter PCB – *Introduction to Electrical Engineering* April 2026
- Simulated a buck converter in LTspice to step down DC voltage and regulate output across varying loads
- Sized the inductor, MOSFET, and diode to balance current ripple, conversion losses, and thermal limits
- Laid out and fabricated a converter prototype to implement the simulated circuit and evaluate DC power delivery

Self-Autonomous Robot – *IEEE - Robotics Automation Society* September 2025 - November 2025
- Built an autonomous maze-navigating robot with color-triggered task transitions and manual-control capability
- Integrated Bluetooth control and a servo launcher to execute ball-targeting tasks in both operating modes
- Won the Best Design Award and placed 2nd of 10 teams at Robotathon, achieving 100% target accuracy

ACTIVITIES

Power Electronics Engineer – Texas Guadaloop January 2026 - Present
- Architect the pod's 600V power system for the linear induction motor and coordinate levitation interfaces.
- Engineer low-voltage distribution and signal connections to integrate electrical subsystems across pod teams.

Procurement Analyst – Texas Guadaloop September 2025 - Present
- Manage $13K+ in across research, business, & engineering teams, aligning specifications and budgets with goals.
- Secure corporate sponsorships through outreach and coordinate purchasing needs with vendor capabilities.

ADDITIONAL INFORMATION

Honors: USA Swimming Scholastic All-American November 2024
UIL Texas Swimming & Diving Academic All-American February 2024, 2025
Interests: Swimming, running, powerbuilding, photography, Filipino cuisine, pickleball, computer building
Work Eligibility: Eligible to work in the U.S. with no restriction